Software Engineering Lab

Group Number – 22

Assignment :- Prepare the risk mitigation plan for the selected project.

Project Details -

Problem Statement:
Enhancing company registration security by implementing real-time camera-based age and gender verification to prevent fraudulent entries.

Company Details -

Organization: Senim Solutions LLP

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Group Members :-

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Objectives of risk mitigation -

- Minimize potential impact
- Enhance probability control
- Ensure project continuity
- Improve stakeholder confidence
- Optimize resource utilization

Metrics used for Risk Management

RISK SEVERITY KEY	RISK LIKELIHOOD KEY	RISK LEVEL KEY	ACCEPTABLE TO PROCEED? KEY
ACCEPTABLE	IMPROBABLE	**LOW**	**YES**
TOLERABLE	POSSIBLE	**MEDIUM**	**NO**
UNDESIRABLE	PROBABLE	**HIGH**	
INTOLERABLE		**EXTREME**	

RISK MATRIX:-

NAME	Enhancing company registration security by implementing real-time camera-based age and gender verification to prevent fraudulent entries.							
REF/ID	PRE-MITIGATION				MITIGATIONS / WARNINGS / REMEDIES	POST MITIGATION		
	RISK	RISK SEVERITY	RISK LIKELIHOOD	RISK LEVEL		RISK LIKELIHOOD	RISK LEVEL	ACCEPTABLE TO PROCEED?
1	Failure in Timely Delivery of Software	UNDESIRABLE	PROBABLE	**MEDIUM**	Proper scheduling of tasks. Team must follow timelines in Gantt chart. Delegation	IMPROBABLE	**LOW**	**YES**
2	User Interface Not Friendly	UNDESIRABLE	PROBABLE	**MEDIUM**	Continuous feedback from user side must be inculcated in software dev elopment life cycle. Focus on user interface during testing phase.	IMPROBABLE	**LOW**	**YES**
3	Data Loss or Corruption	INTOLERABLE	POSSIBLE	**HIGH**	Implement robust data backup and recov ery mechanisms. Regularly test backup systems. Employ encryption and access controls.	IMPROBABLE	**MEDIUM**	**YES**
4	Poor Performance	INTOLERABLE	PROBABLE	**EXTREME**	Conduct performance testing. Optimize code and database queries. Monitor system performance.	PROBABLE	**HIGH**	**YES**
5	Scope Creep	TOLERABLE	PROBABLE	**MEDIUM**	Define and document project scope clearly. Establish a change management process. Communicate proposed changes to stakeholders.	IMPROBABLE	**LOW**	**YES**
6	Inadequate Testing	UNDESIRABLE	POSSIBLE	**HIGH**	Dev elop a comprehensive test plan. Conduct thorough testing at each stage of the dev elopment lifecycle. Utilize automated testing tools.	PROBABLE	**MEDIUM**	**YES**
7	Lack of Stakeholder Engagement	UNDESIRABLE	POSSIBLE	**MEDIUM**	Engage stakeholders proactiv ely. Solicit input and feedback. Address concerns promptly.	PROBABLE	**LOW**	**YES**
8	Deprecated libraries used for the software dev elopment	INTOLERABLE	POSSIBLE	**HIGH**	Keep updating the code and the project in alienation with updates in libraries	PROBABLE	**MEDIUM**	**YES**
9	Integration Issues	UNDESIRABLE	PROBABLE	**HIGH**	Conduct comprehensive integration testing with all relev ant systems and components throughout the dev elopment process. Utilize standardized APIs and protocols to facilitate seamless integration. Engage with third-party vendors early in the project to address any integration challenges proactively.	PROBABLE	**MEDIUM**	**YES**

10	Technical Challenges	TOLERABLE	POSSIBLE	**HIGH**	Conduct thorough technical feasibility studies before project initiation. Allocate sufficient time for prototyping and proof of concept to identify and address potential technical hurdles early on. Engage with experienced technical experts or consultants if necessary.	PROBABLE	**MEDIUM**	**YES**
11	Unclear or Evolving Requirements:	TOLERABLE	PROBABLE	**MEDIUM**	Implement an iterativ e dev elopment approach, such as Agile, allowing for frequent feedback and requirement adjustments	IMPROBABLE	**LOW**	**YES**
12	Data not sav ed properly	INTOLERABLE	PROBABLE	**HIGH**	Utilize transactional database operations to ensure data integrity. Conduct thorough testing of data saving functionality.	IMPROBABLE	**MEDIUM**	**YES**
13	Inventory search results false positiv es	INTOLERABLE	PROBABLE	**HIGH**	Refine search algorithms and criteria to minimize false positiv es. Implement advanced search features such as filters and sorting options. Conduct user testing to validate search results accuracy.	PROBABLE	**MEDIUM**	**YES**
14	Estimation Overruns	ACCEPTABLE	POSSIBLE	**MEDIUM**	Break down project tasks into smaller, more manageable units for more accurate estimation. Incorporate buffer time into project timelines to account for unforeseen delays or scope changes.	IMPROBABLE	**LOW**	**YES**